

Polynomial conditioning for a fixed number of separated Legendre bands

Proof independently reviewed by root, size_bounds, and independent_directions

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All norms in this note are for the uniform probability measure on $[0, 1]$. Put

$$H_d(t) = \int_0^t P_d(2u - 1) du \quad (d \geq 1).$$

Thus $H'_d = P_d(2t - 1)$ and $H_d(0) = H_d(1) = 0$. This note supplies conditioning for a possible fixed-number-of-small-dice construction. It does not by itself construct that family.

Lemma 1. *Let $D \geq d + 1$, p, q be real polynomials, $\deg p < d$ and $\deg(p + H_d q) \leq D$. Then*

$$\|H_d q\|_2 \geq c(D + 1)^{-4} \|q\|_2, \quad (\|p\|_2^2 + \|q\|_2^2)^{1/2} \leq C(D + 1)^5 \|p + H_d q\|_2.$$

Consequently replacing H_d by ηH_d , $0 < \eta \leq 1$, costs at most an additional factor η^{-1} in the second inequality.

Proof. The identity

$$H_d(t) = \frac{(x^2 - 1)P'_d(x)}{2d(d + 1)}, \quad x = 2t - 1,$$

shows that its $d + 1$ simple zeros are the Gauss–Lobatto nodes, including both endpoints. Their positive normalized quadrature weights are

$$\omega_i = \frac{1}{d(d + 1)P_d(2t_i - 1)^2}.$$

They sum to one, and the rule is exact through degree $2d - 1$. These are the classical Legendre–Lobatto formulas; they also follow by integrating the cardinal polynomials and using Legendre orthogonality. In particular,

$$|H'_d(t_i)| \geq [d(d + 1)]^{-1/2} \geq c/d, \quad \|H''_d\|_\infty \leq Cd^2.$$

On an interval of radius cd^{-3} around each zero, its derivative therefore has magnitude at least c/d . These intervals can be made disjoint, by Rolle’s theorem and the same derivative bound. There is exactly one critical point between consecutive zeros, and no others. Thus $|H_d|$ is unimodal on each such interval. For $a = c_0(D + 1)^{-4}$, with sufficiently small absolute c_0 ,

$$|\{t : |H_d(t)| < a\}| \leq Cad^2.$$

The near-zero neighborhoods cover the whole sublevel set because their outer boundary values are at least cd^{-4} , and $D \geq d$. The endpoint neighborhoods are one-sided and satisfy the same estimate.

For a polynomial q of degree at most D , the orthonormal Legendre expansion gives $\|q\|_\infty \leq (D + 1)\|q\|_2$. Hence at most Cc_0 of its squared norm lies in this sublevel set. Taking c_0 small gives the first asserted inequality.

Write $f = p + H_d q$. Since $2 \deg p \leq 2d - 2$, Lobatto exactness gives

$$\|p\|_2^2 = \sum_i \omega_i p(t_i)^2 = \sum_i \omega_i f(t_i)^2 \leq \|f\|_\infty^2 \leq (D+1)^2 \|f\|_2^2.$$

Thus $\|H_d q\|_2 \leq (D+2)\|f\|_2$. The multiplication lower bound now proves the second inequality. Apply it to ηq for the scaled assertion. Degree cancellation cannot hide a larger degree of q , since $\deg p < d < \deg H_d$. $\square$

Theorem 1 (Lacunary product conditioning). *Fix an integer $k \geq 1$ and $r \geq 0$. Choose integers $d_1, \dots, d_k$ so that*

$$d_i > r + \sum_{j < i} (d_j + 1), \quad D = r + \sum_{i=1}^k (d_i + 1).$$

For $0 < \eta_i \leq 1$ and arbitrary polynomials $p_S \in \mathcal{P}_r$, indexed by subsets $S \subseteq [k]$, one has

$$\left(\sum_{S \subseteq [k]} \|p_S\|_2^2 \right)^{1/2} \leq C_k (D+1)^{5k} \left(\prod_{i=1}^k \eta_i^{-1} \right) \left\| \sum_{S \subseteq [k]} p_S(t) \prod_{i \in S} \eta_i H_{d_i}(t) \right\|_2. \quad (1)$$

In particular this representation is unique. Taking $d_i = 1 + r + \sum_{j < i} (d_j + 1)$ gives $D = 2^k(r+2) - 2$. For fixed k , polynomially small η_i therefore produce a polynomial lower bound for the smallest singular value.

Proof. Decompose the displayed sum as $F_0 + \eta_k H_{d_k} F_1$. Both F_0 and F_1 have degree at most $r + \sum_{j < k} (d_j + 1) < d_k$. Lemma 1 bounds the sum of their squared norms by the squared norm of the full sum times $C(D+1)^{10} \eta_k^{-2}$. Apply the same argument recursively to each branch. At most k levels occur, all degrees are at most D , and each level contributes its corresponding η_i^{-1} . This proves (1), including the harmless constant depending only on k . $\square$

Corollary 1 (Products of monotone ranks). *Put $R_i(t) = t - \eta_i H_{d_i}(t)$, with $0 < \eta_i \leq 1/2$. These maps are strictly increasing from 0 to 1. Suppose*

$$d_i > r + k + \sum_{j < i} (d_j + 1), \quad D = r + k + \sum_i (d_i + 1).$$

Then the family of functions

$$\{\phi_a(t) \prod_{i \in S} R_i(t) : 0 \leq a \leq r, S \subseteq [k]\}$$

is linearly independent, and its least singular value in $L^2[0, 1]$ is at least $c_k(D+1)^{-5k} \prod_i \eta_i$. Here ϕ_a are the ordinary orthonormal shifted Legendre polynomials.

Proof. Since $R'_i = 1 - \eta_i P_{d_i}(2t-1) \geq 1/2$, monotonicity is immediate. For $F = \sum_S p_S \prod_{i \in S} R_i$, expansion gives

$$F = \sum_T (-1)^{|T|} \left(\prod_{i \in T} \eta_i H_{d_i} \right) q_T, \quad q_T = \sum_{S \supseteq T} t^{|S \setminus T|} p_S.$$

Each q_T has degree at most $r + k$. The inverse triangular identity is

$$p_S = \sum_{T \supseteq S} (-t)^{|T \setminus S|} q_T.$$

As multiplication by a power of t is an L^2 contraction, both transformations have norm bounded by a constant depending only on k . Theorem 1, applied with degree budget $r + k$, proves the claim. $\square$

Corollary 2 (Discrete sampling). *For the spaces in Theorem 1 or Corollary 1, the same conditioning estimates, up to an absolute factor, hold in empirical L^2 on any node set with star discrepancy at most $c(D+1)^{-2}$. In particular they hold on each of a fixed number of interlaced groups of a sufficiently fine perturbed midpoint grid.*

Proof. Every function F involved is a polynomial of degree at most D . The discrepancy inequality and the L^2 Markov inequality give

$$\left| \frac{1}{N} \sum_q F(t_q)^2 - \int_0^1 F^2 \right| \leq \text{disc}(t) \int_0^1 |(F^2)'| \leq C(D+1)^2 \text{disc}(t) \|F\|_2^2.$$

For sufficiently small c this is at most half the squared norm. One elementary proof of the needed Markov inequality expands F in Legendre polynomials and uses $\|\phi'_j\|_2^2 = 2(2j+1)j(j+1)$, followed by Cauchy–Schwarz. $\square$